Budget Guard: Calibrated Resource Control for Long-Horizon LLM Agents

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Abstract

Long-horizon large language model (LLM) agents can consume substantial tokens, time, and money before recognising that a task is unlikely to succeed. Prior work on budget-aware agents shows the promise of early stopping, but agents' own remaining-budget estimates are poorly calibrated, so they cannot be trusted for resource-control decisions. We introduce Budget Guard, a model-agnostic framework that wraps an unmodified agent in a conformal calibration layer and requires no fine-tuning of the underlying LLM. It combines feature-dependent prediction intervals with adaptive and sequential conformal methods to handle temporal dependence and deployment shift, and extends calibration to several jointly constrained resources and to censored runs. On Claude Opus 4.7 rollouts from Sokoban, Search-R1 and SWE-bench, offline split-CQR reaches 0.90–0.96 empirical coverage and online windowed-CQR reaches 0.87–0.92 against a nominal 0.90, where the agent's own intervals reach 0.51, 0.23 and 0.35; across all 19 model–environment runs, on one split each, windowed-CQR holds at least 0.85 coverage in 18 of 19. In the multi-resource Warehouse environment, in-distribution Bonferroni calibration reaches 0.95 joint coverage across time, capacity and cost, though it falls to 0.71 under a model swap. Thresholding a separately calibrated feasibility score at a false-abort tolerance $\delta = 0.10$ avoids 86.3% of the spend on doomed runs with no false aborts observed, and cuts estimated Cost-of-Pass from \$1.61M to \$0.87M in simulated Warehouse dollars per successful run at full pass retention — all in counterfactual replay of logged rollouts, not live deployment. Reliable budget awareness can therefore be supplied as an external statistical control layer, without additional model training.

1 Introduction

LLM agents increasingly complete long-horizon tasks that require many reasoning steps, tool calls, and interactions with external environments [1], and those capabilities carry growing resource demands [2]. An agent may consume thousands of tokens, substantial wall-clock time, or real money before finishing a task [3], and it may spend a good deal of that on redundant reasoning that does not improve the outcome [4–6]. When a trajectory eventually fails, most of that spending buys nothing [3]. Controlling resource consumption without giving up task performance is therefore a practical obstacle to deploying agentic systems at scale [7–9].

One difficulty is that an agent does not reliably know how much budget it still needs [3]. It can produce an estimate, but those estimates are uncalibrated, so they cannot be trusted for resource-control decisions.

This motivates Budget Guard, a model-agnostic framework for calibrated resource control in long-horizon LLM agents. Rather than teaching the underlying LLM budget awareness through fine-tuning,

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Budget Guard operates as an external statistical layer around an existing agent. At each trajectory prefix it uses observable information — resources consumed so far, progress, recent usage patterns, and the agent's own estimate when one is available — to predict the resources still required for successful completion. No LLM is trained or fine-tuned; the only fitted components are lightweight gradient-boosted trees over usage-accounting features.

The central component is conformal prediction, a distribution-free framework for constructing prediction intervals with a specified coverage target. At a nominal coverage of 90%, the objective is for the true remaining requirement to fall inside the interval about 90% of the time under the relevant calibration conditions — exchangeability offline, or rollout-safe online evaluation when labels arrive only at trajectory completion. Budget Guard starts from Conformalized Quantile Regression (CQR) [10] for feature-dependent intervals, and adds online methods — Adaptive Conformal Inference (ACI) [11], dynamically tuned ACI (DtACI) [12], and a windowed CQR variant — to absorb temporal dependence and distribution shift. This matters most when the underlying LLM changes or when consumption behaviour differs between calibration and deployment.

Our contributions are fourfold. (1) We introduce a conformal framework for calibrating progressive remaining-budget estimates that requires no LLM fine-tuning (§3, §5.1; Table A2). (2) We evaluate adaptive and sequential calibration under environment and model shift, where frozen calibration collapses from 0.92 to 0.10 coverage and online recalibration recovers much of it (§5.2; Figure 3). (3) We extend budget awareness to jointly constrained resources and censored trajectories (§3, §5.3, §5.4; Table 2, Tables B3–B4). (4) We connect a calibrated feasibility signal to an economic stopping policy that avoids 86.3% of doomed-run spend at a bounded false-abort rate (§5.5; Figure 4).

Budget Guard treats budget awareness not only as a prediction problem but as a control problem: estimate what remains, and use the uncertainty in that estimate to decide whether continued execution is worth paying for.

2 Related work

BAGEN [13] formalises the estimation of an agent's remaining budget as *budget awareness* and evaluates whether an agent can estimate its remaining requirements mid-execution. A point estimate says nothing about its own uncertainty, and an interval is only useful if its coverage can be trusted; BAGEN takes intervals directly from the agent, and they are uncalibrated. Conformal prediction, by contrast, constructs intervals with specified coverage guarantees without distributional assumptions [14, 15]. BAGEN also asks the agent to judge, at different prefixes, whether successful completion is still feasible, and covers internal resources such as tokens alongside external ones such as time, money, and warehouse capacity across Sokoban, Search-R1, SWE-bench, and Warehouse.

The BAGEN results expose a gap between task capability and budget awareness: strong agents solve hard tasks while remaining poorly calibrated about what finishing will cost. BAGEN reports that early stopping on budget-awareness signals saves roughly 28–64% of tokens on failed trajectories, at a cost of roughly 1.6–4.2 percentage points of task success, with interval coverage reaching only 47% even after supervised fine-tuning and reinforcement learning [13]. The agent evidently carries some signal about future requirements, but its raw estimates are not reliable enough for high-confidence control.

Doomed From the Start [16] monitors a model's internal signals to decide when a trajectory is doomed and terminates it early. Like us and unlike BAGEN, it applies a distribution-free calibrated abort gate with no fine-tuning, meeting recall targets of 90–97% on TextCraft and saving 47.1% (Qwen-2.5-7B) and 37.2% (Llama-3.2-3B) of inference compute at the 90% target. Because it reads internal states, it applies only to open-weight models.

A third line applies conformal prediction over an agent's *answer*. MiCP [17] allocates an error budget to each turn of a multi-turn reasoning process, letting an agent stop once it has gathered enough evidence while preserving an overall coverage guarantee. MiCP and Budget Guard share the machinery — conformal prediction driving an early-stopping decision in a multi-turn agent — but calibrate different objects: MiCP calibrates coverage over the answer to decide when to stop reasoning, while Budget Guard calibrates coverage over the remaining resource budget to decide whether a run is worth continuing. Budget Guard is also an external wrapper rather than a component of the reasoning loop, and additionally handles jointly constrained resources, censored trajectories,

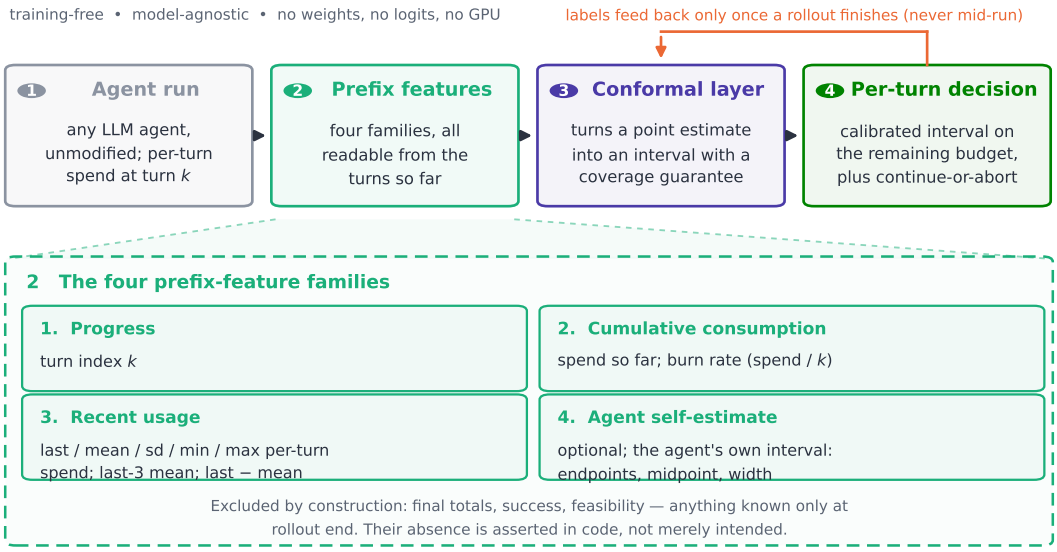


Figure 1: **Budget Guard wraps an unmodified agent in an external calibration layer.** *Input:* per-turn usage accounting only — the tokens, dollars, or time a provider API already reports, plus the agent's self-reported interval when it emits one. *Output:* a calibrated interval on the remaining budget and a continue-or-abort decision, at every turn. *Key component:* the conformal layer (stage 3), which converts a point estimate into an interval carrying a coverage guarantee. Two properties make the setting unusual and are drawn explicitly: the guard reads no model weights, logits, or environment internals, and a label becomes observable only when a rollout ends, so online recalibration may never use a still-running rollout.

and an economic stopping policy. Appendix A defines the background concepts used to interpret our results.

3 Method

Setup. At turn k of a rollout, Budget Guard observes only the prefix: per-turn resource usage so far and, when available, the agent's own estimate of its remaining budget. It outputs an interval for y, the resources the run still needs to finish successfully, at nominal miscoverage $\alpha = 0.10$. Two properties of this label shape the design. First, y is observable only once the rollout finishes, so no online method may consume a label mid-run. Second, y is defined only while the run can still finish: a doomed run censors its label rather than delaying it. The guard acts as a verifier — it need not trust the agent's self-report, because it wraps that self-report in a guarantee that holds regardless of its quality (Table A1 states the guarantee each component carries). Figure 1 gives the core loop: prefix features feed a conformal predictor that returns a calibrated interval on the remaining budget. The rest of this section extends that loop three ways — offline and online calibration variants for deployment shift, joint calibration when several resources bind at once, and censoring-aware bounds behind a feasibility gate for runs that can no longer finish — and then turns the feasibility signal into an economic stopping rule with a per-run false-abort guarantee.

Prefix features. We use four families of features, all computable at turn k with no access to the rest of the trajectory. Three come from the usage stream alone: progress (the turn index k); cumulative consumption (spend so far, and the burn rate spend$/k$); and recent-usage statistics (the last, mean, standard deviation, minimum and maximum per-turn spend, the mean of the last three turns, and the difference between the last turn and the mean). The fourth is the agent's self-estimate — interval endpoints, midpoint, and width — when the agent produces one. Features that depend on the rollout's future, such as final totals, success, or feasibility, are excluded from the feature matrix and asserted absent in code rather than merely left out, so no label information can leak into a prediction. The guard needs no model weights, logits, or environment internals, because every feature comes from usage accounting that provider APIs already report.

Calibrated intervals under deployment shift. The base predictor is a gradient-boosted quantile regressor [18]. We split calibration and test by rollout, never by prefix: prefixes of one rollout share its final total, so splitting them apart would leak each test rollout's answer into calibration. This frozen split-CQR holds only under exchangeability, which breaks under a model swap or a new task type. We therefore also recalibrate from the deployment stream's own finished rollouts, three ways: ACI, which raises or lowers the working coverage level from recent misses [11]; DtACI, which removes ACI's hand-set step size [12]; and windowed-CQR, a conformal quantile over a sliding window of the 100 most recent scores, inspired by SPCI [19]. Its coverage gap grows with window-to-stream drift [20]. All observe labels rollout-safely: a rollout's labels enter the update state only after it finishes, and the state is frozen within a rollout. Revealing them immediately would use labels that do not yet exist at deployment and would inflate coverage, so we report only this deployment-faithful protocol. Two reporting rules follow. An unbounded upper endpoint covers everything and informs nothing, so we report the fraction of unbounded intervals beside every coverage number. And because a single global quantile is set by the many easy late-turn prefixes, a good average can hide under-coverage at the first turns, where an abort decision has the most leverage; we therefore also calibrate a separate conformal quantile per turn-depth bucket (Mondrian [21]): $k = 1$, $k = 2$, $k \geq 3$.

Joint multi-resource calibration. When a run is constrained by several budgets at once (Warehouse: weeks, warehouse item-weeks, and dollars), the relevant event is joint — the run survives only if all three hold — and covering each at level α does not cover their conjunction, because the misses accumulate. Calibrating each resource at $\alpha/3$ (Bonferroni) restores a joint guarantee of $1 - \alpha$ by the union bound, at the price of wider margins; this is our headline multi-resource method. A tighter alternative shares one threshold by ranking the resources' errors jointly [22], but that rank is chosen on the same scores it then thresholds, so it carries no guarantee; we report it as an observation and rest the claim on Bonferroni. Under a model swap the joint calibration also moves: a rolling window of recent three-resource errors updates both the margins and the shared rank, again only once a run has finished.

Censored rollouts and a selective guarantee. When a run is doomed we never learn y, only that it exceeded what was left — the label is censored — and calibrating on uncensored runs alone drops exactly the hard cases. We calibrate one-sided upper bounds five ways: treating censored scores as $+\infty$ (valid, but vacuous once the censored fraction exceeds α); substituting the known lower bound; dropping censored scores entirely (the naive comparison); reweighting by each run's censoring probability (IPCW [23, 24]); and charging a censored run only against the resource that stopped it. A Kaplan–Meier estimator [25] is a diagnostic rather than a sixth option: it reports when the censored upper tail leaves no finite valid bound at all. A feasibility gate — a classifier on the same prefix features — then issues intervals only on predicted-feasible prefixes, so coverage holds among the prefixes it admits rather than across all prefixes. This selective guarantee is what the construction supports; marginal coverage over all prefixes is reported as an empirical observation.

Economic stopping. The abort rule thresholds feasibility: stop at the first turn whose feasibility score falls below τ, the abort threshold. Because the rule aborts a rollout exactly when that rollout's minimum prefix score crosses τ, we calibrate at the rollout level, setting τ to the δ-quantile (with the finite-sample correction) of the minimum scores of feasible calibration rollouts. This bounds by δ — the false-abort tolerance — the probability of aborting a run that would have finished: a per-run guarantee matched to the sequential decision, not a per-prefix statement. Aborting a doomed run recovers its remaining spend, while aborting a feasible run forfeits its pass; both effects enter Cost-of-Pass [26], total spend per passing run, so the rule is evaluated in economic terms rather than as a coverage statistic. The rule is a calibrated binary gate in the sense of conformal risk control [27]: *it thresholds the feasibility score and does not read the remaining-budget intervals*. An interval-driven variant that aborts when spend-so-far plus the calibrated lower bound exceeds the budget is implemented in the released code; all economic results below come from the feasibility-gate rule, and we do not claim the two are equivalent.

4 Experimental setup

Data and environments. All experiments use logged rollouts from BAGEN [13] at a pinned dataset revision; Table 1 summarises the four environments. Sokoban, Search-R1 and SWE-bench are single-resource token tasks. Sokoban and Search-R1 use uniform context budgets of 2,500 and 3,500 tokens; on SWE-bench the official budget varies by run (6,772, 7,060, 8,265, 10,472 and 21,711

Table 1: **The four evaluation environments; only Warehouse binds several resources at once, and only Warehouse censors labels.** Token budgets are the BAGEN context budgets in tokens; Warehouse budgets are set per task. In Warehouse, half of the prefixes (640 of 1,280 per model) belong to runs that can no longer finish and drive the censored-calibration evaluation.

Environment	Task	Budgeted resource(s)	Budget	# Models
Sokoban	box-pushing puzzle	tokens	2,500 tokens	7
Search-R1	multi-turn retrieval QA	tokens	3,500 tokens	7
SWE-bench	repository-level bug fixing	tokens	6,772–21,711 (per run)	5
Warehouse	multi-week management sim.	time, capacity, cost	per task, joint	5

tokens across the five models), and we read each run's value from its estimation record rather than applying a per-environment constant. The label is the number of tokens still needed to complete the task under BAGEN's turn-excluding-history accounting, which our extractor reproduces exactly against the official per-sample ground truth (0 of 355 mismatches on the reference run). Warehouse is a multi-week management simulation with three jointly binding budgets — time in weeks, capacity in item-weeks, and cost in simulated USD — and per-prefix ground truth for all three comes from the official estimation records. Half of its prefixes (640 of 1,280) belong to runs that can no longer finish, which makes Warehouse our censoring test.

Protocol and baselines. Rollouts are split 50/25/25 into train, calibration and test sets. We repeat every headline experiment with five random splits (seeds 0–4), assigning whole rollouts to a split rather than individual prefixes, so no rollout has its prefixes divided across sets; reported coverage and economics are means over the five seeds. Online methods consume the test stream in its natural order with labels revealed only at rollout boundaries. We evaluate three shift scenarios: cross-environment (calibrate on Sokoban, deploy on Search-R1 and SWE-bench), model swap (calibrate on Claude Opus 4.7, deploy on four other agents' SWE-bench streams), and an external-resource model swap on Warehouse. Seven comparators run on the same splits and seeds as the guard: the agent's own interval, a feature-free constant-quantile interval, split conformal on absolute residuals, a conformalized-agent interval, a guard-plus-self-estimate ablation, frozen split-CQR under shift, and hard caps for stopping. Coverage is reported against the 0.90 nominal level, always alongside the fraction of unbounded intervals. Appendix C gives the full protocol, baseline definitions, metric formulas, and implementation details.

Models. We evaluate seven agents: Claude Opus 4.7, Claude Opus 4.6, Claude Sonnet 4.6, GPT-5.2-instant, GPT-5.2-Codex, Gemini 3.1 Pro, and Qwen3-235B. Crossed with environments this gives 19 token-budget model$\times$environment runs (7 Sokoban, 7 Search-R1, 5 SWE-bench) of 52–128 rollouts and 85–3,715 prefix samples each; per-run counts are in Table B1, Appendix B.

5 Results

We evaluate prediction intervals using empirical coverage, normalised width, and the fraction of unbounded intervals. Unbounded intervals raise coverage while carrying no information, so we hatch them in the figures and report the unbounded fraction beside every coverage number.

5.1 Within-run interval calibration

Figure 2 shows within-run remaining-budget calibration on Sokoban, Search-R1 and SWE-bench using Claude Opus 4.7 rollouts. Agent self-reported intervals are badly calibrated, covering only 0.51, 0.23 and 0.35 of prefixes against a 0.90 target. Offline conformal calibration largely closes that gap, with split-CQR reaching 0.92, 0.90 and 0.96. Our online windowed-CQR reaches 0.92, 0.92 and 0.87 with no unbounded intervals in any of the three environments. ACI-CQR and DtACI sometimes reach higher coverage, but do so partly by emitting unbounded intervals on up to 16% of the stream, which shows that coverage alone is not a sufficient criterion. Across all 19 logged model–environment runs, windowed-CQR reaches at least 0.85 coverage in 18 of 19 and at least 0.90 in 11 of 19, against 14 and 8 for DtACI, on one split per run (Table B1). It is also sharper than the feature-free baseline on two environments of three: 22% narrower than constant-quantile on Sokoban and 55% narrower on SWE-bench, but 23% wider on Search-R1, where constant-quantile is the sharpest bounded method

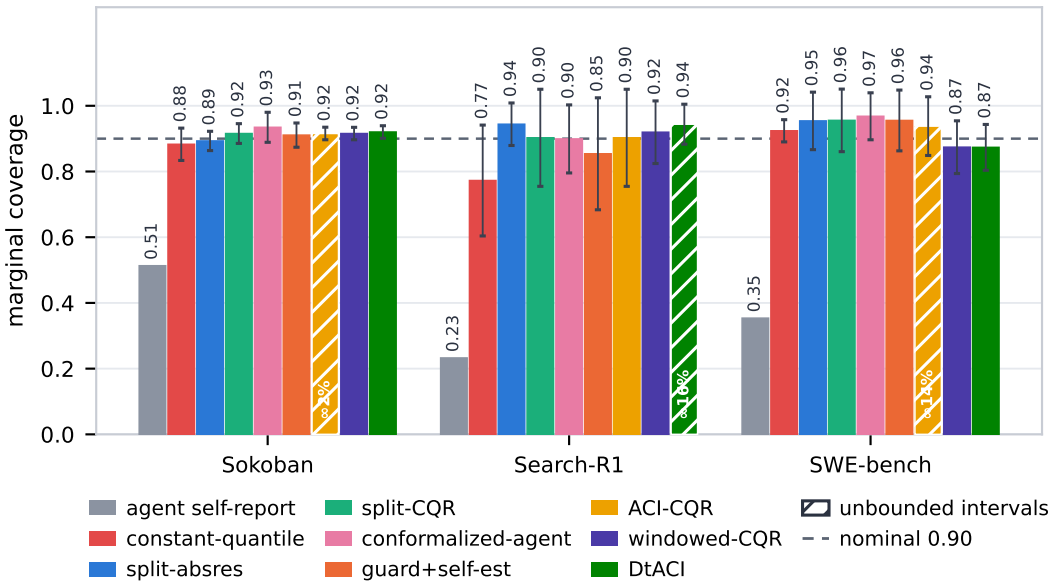


Figure 2: **Conformal calibration closes almost all of the gap between agent self-reports and the 0.90 target, and windowed-CQR does it without ever giving up and returning an unbounded interval.** Marginal coverage of the true remaining budget on Claude Opus 4.7 rollouts, $\alpha = 0.10$, averaged over five rollout-level split seeds; error bars are the standard deviation across those seeds. Higher is better up to the nominal 0.90 (dashed). Hatched bars emit unbounded intervals on the annotated fraction of the stream, which buys coverage at the cost of informativeness. Agent self-report is a single value read from the logged records rather than a seed mean. Coverage, normalised width, unbounded fraction and test-set sizes for all methods are in Table A2.

(Table A2). Adding the agent's own estimate to the guard's features does not help — guard+self-est is within 0.01 of split-CQR on Sokoban (0.91 against 0.92), identical on SWE-bench (0.96), and worse on Search-R1 (0.85 against 0.90) — consistent with the self-reports carrying little signal the usage stream does not already contain.

5.2 Robustness under deployment shift

Figure 3 evaluates Budget Guard under deployment shift. Transferring a guard calibrated on Sokoban to Search-R1 and SWE-bench drops frozen split-CQR coverage to 0.51 and 0.10. Rollout-safe windowed-CQR partially recovers it, to 0.71 and 0.80, with no unbounded intervals; ACI and DtACI reach higher coverage but need unbounded intervals on 60–88% of predictions to get there. In the model-swap setting a guard calibrated on Claude Opus 4.7 is transferred to Sonnet, Gemini, GPT and Qwen SWE-bench streams. Windowed-CQR reaches 0.66–0.89 there, and its seed-to-seed standard deviation is an order of magnitude smaller than frozen calibration's (0.006–0.011 against 0.122–0.224), so its failures are at least predictable. Online adaptation improves robustness under shift, but severe shift remains unsolved: no bounded-interval method recovers the nominal level, and high coverage has to be read together with interval informativeness.

5.3 Conditional and multi-resource calibration

A good marginal number can hide a bad conditional one. On Sokoban, split-CQR reaches 0.92 marginal coverage but only 0.86 at the first turn, exactly where an abort decision has the most leverage; Mondrian calibration by turn depth raises first-turn coverage to 0.93. On Search-R1 and SWE-bench, where the marginal-to-first-turn gap is already small, Mondrian is neutral to slightly negative (Table B2), so it buys conditional validity rather than a uniform improvement.

Table 2 reports the multi-resource case. In Warehouse, agent self-reports reach 0.42 joint coverage across time, capacity and cost. Calibrating each resource independently at α lifts the joint event only to 0.80, which is the union bound biting: three margins at 0.90 do not make a conjunction at 0.90.

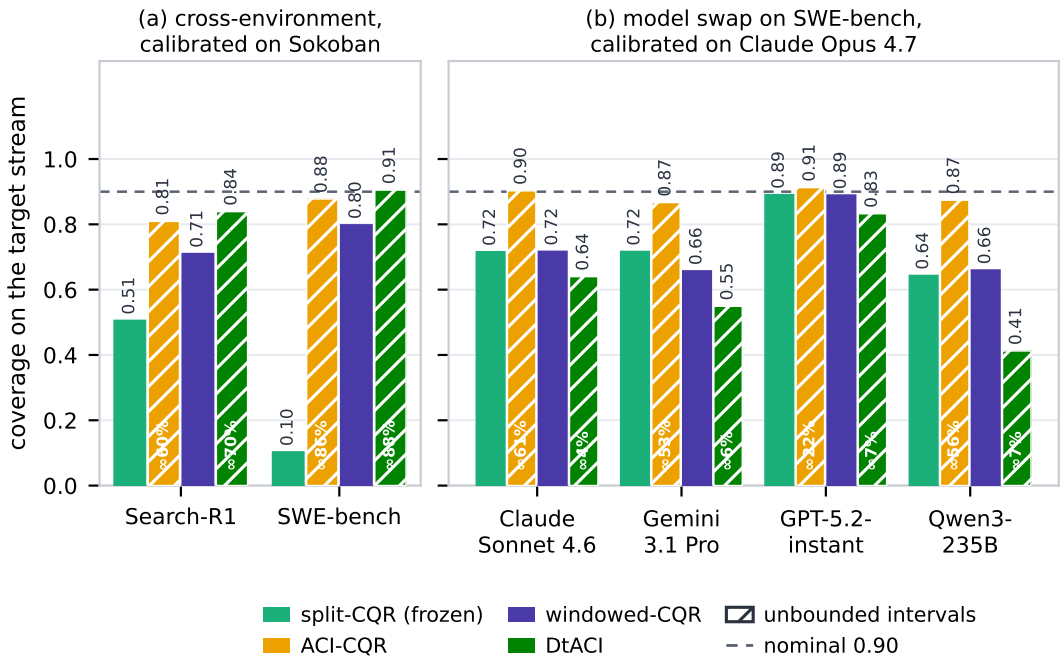


Figure 3: **An environment change collapses frozen calibration and online recalibration recovers much of it; a model swap defeats every method we tried.** Marginal coverage on the target stream; higher is better up to the nominal 0.90 (dashed). **(a)** A guard calibrated on Sokoban and deployed unchanged on Search-R1 and SWE-bench: frozen split-CQR falls to 0.51 and 0.10, and windowed-CQR recovers to 0.71 and 0.80. **(b)** A guard calibrated on Claude Opus 4.7 SWE-bench rollouts and deployed on four other agents: here frozen and online calibration are comparable on average (0.74 against 0.73) and neither reaches the target, though windowed-CQR varies far less across seeds. Hatched bars emit unbounded intervals on the annotated fraction of the stream — 60–88% for the ACI-based methods in (a), and 22–61% (ACI-CQR) and 4–7% (DtACI) in (b).

Table 2: **Independent margins under-cover the joint event, Bonferroni fixes it in distribution, and nothing with a guarantee survives a model swap.** Warehouse coverage against a nominal 0.90, mean over five rollout-level split seeds; higher is better up to 0.90, above which coverage is bought with wider margins. Per-resource columns and the in-distribution joint column are Claude Opus 4.7; the final column is the joint event averaged over four swap targets. Bold marks the highest joint coverage in each column; note that Bonferroni's excess over 0.90 in distribution is the union bound being conservative, not extra accuracy. "–" marks a scheme not run in that setting.

	per-resource coverage			joint coverage	
Calibration scheme	time	capacity	cost	in-distribution	model swap
agent self-report	0.87	0.67	0.71	0.42	0.34
independent margins (α)	0.99	0.89	0.92	0.80	–
Bonferroni ($\alpha/3$)	1.00	0.98	0.97	**0.95**	0.71
empirical copula	0.99	0.94	0.96	0.89	0.62
adaptive copula (online)	–	–	–	–	**0.87**

Bonferroni at $\alpha/3$ restores 0.95, and the empirical copula reaches 0.89 with narrower margins but no guarantee. Under an external-resource model swap, however, frozen Bonferroni falls to 0.71 — below its nominal level, and below the 0.87 reached by the un-guaranteed adaptive copula, which recalibrates from finished rollouts. The guaranteed method is the right default in distribution and the wrong one under shift; we report both rather than choosing the flattering one.

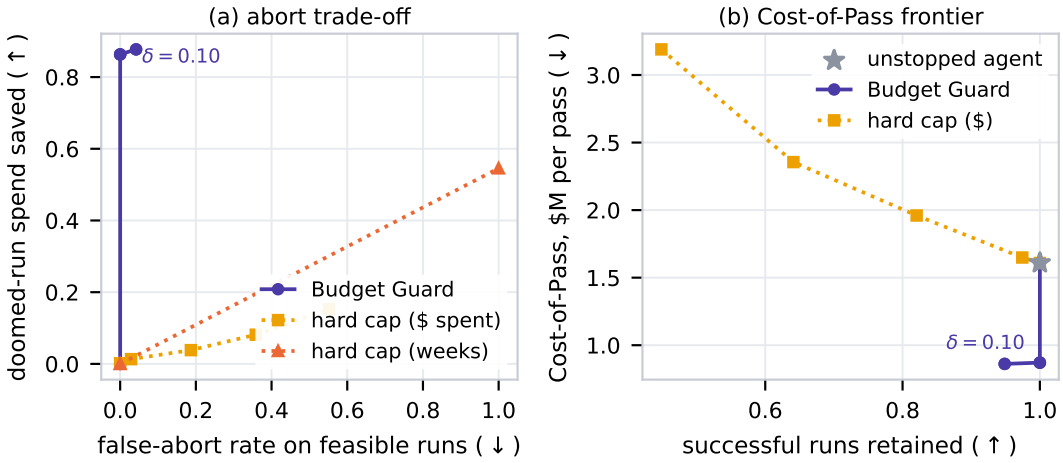


Figure 4: **At its $\delta = 0.10$ operating point the calibrated gate avoids 86.3% of doomed-run spend with no observed false aborts, which no hard cap achieves at any setting.** Warehouse, simulated USD, mean over five seeds. **(a)** Every hard cap that saves more than 0.2% of doomed spend also stops runs that would have succeeded; the guard sits at the top-left corner instead. **(b)** Cost-of-Pass against the fraction of successful runs retained: the guard roughly halves cost per pass while retaining every observed success, whereas no cap beats the unstopped agent — discarding passes raises cost per pass, to $3.19M at the tightest cap. Both panels are counterfactual replays of logged rollouts, not live deployments.

5.4 Censored calibration

Roughly half of Warehouse prefixes are censored: the run can no longer finish, so its true completion requirement is never observed. Dropping those observations yields only 0.50 coverage on the censored prefixes where cost is the binding resource, so failed trajectories cannot be treated as missing data. Treating censored scores as $+\infty$ is valid but vacuous at this censoring rate, and the informative treatments trade validity for sharpness in ways Table B4 makes explicit. Feasibility-gated selective intervals offer the other lever: abstaining on predicted-doomed prefixes holds coverage at 0.92–0.95 for capacity and cost, and at 1.00 for time, on the prefixes it keeps, at the cost of keeping between 51% and 89% of them (Table B3). The gate is swept over its own prefix-level tolerance δ_{gate}, which is distinct from the rollout-level δ of §5.5. One caveat governs how these numbers should be read: the recovered labels used to score censored prefixes are exact only for time, and inherit a week-proportional interpolation with 12–27% median relative error for capacity and cost, so those two are coverage against an approximate label rather than ground truth.

5.5 Economic impact

Figure 4 evaluates early stopping in Warehouse. At $\delta = 0.10$ the calibrated feasibility gate avoids 86.3% of doomed-run spend with 0% observed false aborts. Hard spending caps achieve almost nothing at zero false aborts — the tightest cap that preserves every successful run saves 0.2% of doomed spend — and dollar caps reach meaningful savings only at false-abort rates of 19–55%; the only week cap that saves anything (54.5%) aborts every feasible run. In the Cost-of-Pass analysis the guard reduces estimated cost per successful trajectory from $1.61M to $0.87M, a reduction of 46%, while retaining all observed successful runs. No hard cap beats the unstopped agent by more than a rounding error: discarding passes *raises* cost per pass, to $3.19M at the tightest cap. Two cautions. This 86.3% is simulated Warehouse dollars saved on doomed runs, and is not comparable to BAGEN's 28–64% token savings on failed trajectories, a different resource in different environments. And the operating point rests on a small number of test rollouts per seed, so the observed 0% false-abort rate should be read as consistent with the $\delta = 0.10$ bound rather than as evidence that the true rate is zero.

6 Discussion and limitations

Our joint results rest entirely on Warehouse. Bonferroni calibration reaches 0.95 joint coverage there, but a single multi-resource environment cannot support a claim that the approach generalises, and §5.3 shows that guarantee is not robust to a model swap. Our breadth figures are single-split counts rather than seed averages: on seed 0, windowed-CQR reaches at least 0.85 coverage in 18 of 19 model×environment runs, but not all 19, and DtACI in only 14. Even where average coverage meets the target, an individual run can be badly miscalibrated, and marginal coverage can hide under-coverage in slices such as early turns or particular resources. Some test splits are small: Search-R1 leaves about 24 test prefixes per seed, which is why its standard deviations reach 0.10–0.17 and why several Search-R1 entries in Table B1 sit at exactly 1.000. Those numbers are indicative rather than precise.

Results are also method-dependent, and the guarantee is a promise from the mathematics that holds only while its assumptions do. Conformal methods are built to hit their coverage target and to widen intervals exactly as much as that requires, so coverage alone does not separate a useful interval from a trivially wide one; we therefore report width and unbounded fraction throughout. Our intervals are roughly 22% narrower than a constant-quantile baseline on Sokoban and 55% narrower on SWE-bench, though 23% wider on Search-R1; and windowed-CQR holds up under shift with no unbounded intervals while the adaptive variants become unbounded on a large share of shifted runs — though, as §5.2 shows, windowed-CQR still under-covers the nominal level under severe shift. Our censored results carry a further caveat: for capacity and cost they are measured against an interpolated label with 12–27% median relative error, not against ground truth.

Finally, the economic results are estimates rather than live measurements. The coverage numbers are measured directly: an interval either holds the true budget or it does not. The savings and Cost-of-Pass figures are obtained by replaying completed runs, cutting them off where the guard would have aborted, and summing the spending avoided. That is a fair estimate, since each run's real outcome and cost are known, but it assumes that adding the guard would not change how the agent behaves and that our sample of runs is representative. The 86.3% savings and the \$1.61M-to-\$0.87M Cost-of-Pass reduction are best read as what the guard would likely have saved, not as numbers from running it live.

Broader impact. The intended effect is to reduce wasted compute, energy and money on agent runs that cannot succeed. The corresponding risk is that a stopping threshold set too aggressively terminates runs that would have finished; the δ calibration in §3 exists precisely to bound that risk at a user-chosen level, and the counterfactual caveat above means a deployer should not treat the reported savings as a guaranteed live return when choosing δ.

7 Conclusion

Budget Guard shows that reliable budget awareness can be supplied as an external control layer for long-horizon LLM agents, with no fine-tuning of the underlying model. Applying conformal calibration to progressive remaining-budget estimates yields 0.90–0.96 empirical coverage offline (split-CQR) and 0.87–0.92 online (windowed-CQR) against a 0.90 target on Claude Opus 4.7 rollouts, and holds at least 0.85 coverage in 18 of 19 model×environment runs, and extends to several jointly constrained resources, though the joint guarantee does not survive a model swap. Separately, a calibrated feasibility gate supports early-stopping decisions that cut spending on runs unlikely to succeed while bounding the risk of terminating successful ones: in counterfactual replay at the Warehouse $\delta = 0.10$ operating point it avoids 86.3% of doomed-run spend with no observed false aborts and reduces estimated Cost-of-Pass from \$1.61M to \$0.87M in simulated Warehouse dollars per successful trajectory, retaining every observed success. Budget awareness can therefore serve not only as a measurement but as a control signal during execution, which is a practical step toward agentic systems that are cheaper to run without being less capable.

References

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A Background and guarantees

This appendix defines the concepts needed to interpret the results.

- **Coverage.** The fraction of cases in which the true remaining budget falls inside the predicted interval. Coverage of 0.90 means the true value is contained in about 90% of cases.
- **Calibration set.** A held-out set of completed trajectories used to set interval widths and stopping thresholds.
- **Rollout (run).** One complete agent attempt at a task.
- **Turn (k).** One agent step within a rollout.
- **Prefix.** The partial trajectory containing all turns completed through turn k.
- **$\hat{q}$.** The conformal threshold: a learned error margin added to the model's predictions to expand them into an interval. Throughout, $\hat{q}$ is the conformal threshold and $\hat{f}_{\mathrm{lo}}, \hat{f}_{\mathrm{hi}}$ are the quantile regressor's own predictions.
- **α.** The allowed miscoverage rate; the target coverage is $1 - \alpha$. We use $\alpha = 0.10$ throughout. A smaller α raises the target and generally widens intervals.
- **Width.** The size of the predicted range. Narrower is more informative at equal coverage: $[100, 115]$ is more useful than $[100, 300]$. An interval extending to infinity attains coverage while saying nothing.
- **δ.** The false-abort tolerance for the stopping rule. A larger δ permits greater savings on doomed runs while accepting more risk of aborting successful ones.
- **τ.** The abort threshold. The guard aborts at the first turn whose feasibility score falls below τ. It is not chosen by hand: it is read off the calibration rollouts as the value holding the false-abort rate at or below δ, so a smaller δ produces a more cautious τ.
- **Offline.** Calibrates once on completed runs and stays fixed during deployment.
- **Online.** Updates calibration after each completed run, using its newly observed remaining-budget labels.

Table A1 states, component by component, what is guaranteed and under what assumption. Four rows carry no finite-sample guarantee at all — windowed-CQR, the empirical copula, the censoring-aware bounds other than $+\infty$, and the marginal rather than selective reading of the gated intervals — and every claim resting on those is reported in this paper as an empirical observation.

B Per-run breadth results

Table B1 lists every model×environment run behind the breadth claims in §5.1; the "18 of 19" and "11 of 19" counts can be read straight off the windowed-CQR column. Table B2 gives conditional coverage by turn depth, Table B3 the feasibility-gate sweep, and Table B4 the censored upper-bound calibrations. Every number is read from the generated result tables, which one command regenerates and two checkers verify.

C Protocol, baselines, metrics, and implementation

This appendix records what §4 summarises: how the data is split, what each baseline is, how each reported number is computed, and the settings under which all of it runs. Every value below is fixed in the released code, and the parenthetical names are the modules that implement it.

C.1 Splits and seeds

Every split is by rollout, never by prefix (`split.group_split`). For a given seed the unique rollout indices are permuted with `numpy.random.default_rng(seed)` and allocated 50 / 25 / 25 to train, calibration and test; every prefix follows its rollout. Prefixes of one rollout share that rollout's final total, so splitting them apart would place each test rollout's answer into calibration. A split that is requested but comes out empty raises an error rather than degrading quietly: with an empty calibration set the conformal quantile of zero scores is $+\infty$, every interval becomes $[0, \infty)$, and coverage reads a meaningless 1.00 with no other symptom. Headline numbers are means over five seeds, 0 through 4.

Table A1: **What each component guarantees, and what it does not.** Statements we do not have are listed as such, so a reader can tell a theorem from an observation.

Method	Role in Budget Guard	Guarantee and scope
split-CQR	CQR intervals conformalized on a by-rollout calibration split; our offline default [10].	Marginal coverage $\geq 1-\alpha$ over a random (rollout, prefix) pair, under exchangeable rollouts. Pooling a rollout's prefixes is an empirical evaluation, not the guaranteed object [15].
Mondrian-CQR	split-CQR calibrated separately per turn-depth bucket, so early turns are not masked by easy late-turn prefixes [21].	Coverage $\geq 1-\alpha$ within each bucket, under exchangeability within the bucket; thin buckets fall back to the global quantile.
ACI / DtACI	Adaptive coverage levels recalibrated from the stream's finished rollouts; DtACI drops ACI's step-size knob via expert aggregation [11, 12].	Long-run average coverage approaches $1-\alpha$. May emit unbounded intervals; we clamp the working level at 0.999 rather than emit empty sets, a conservative deviation.
windowed-CQR	Conformal quantile over a sliding window of the 100 most recent scores; an SPCI-inspired simplification [19].	No finite-sample guarantee; coverage gap bounded by window-to-stream drift [20]. Evaluated empirically.
Bonferroni joint	Calibrates each resource at $\alpha/3$ so all hold at once; our headline multi-resource method.	Joint coverage $\geq 1-\alpha$ by the union bound; no data-driven selection; conservative, so realised coverage exceeds the target in distribution.
empirical copula	Shared copula rank across resources, tighter than Bonferroni; an adaptive variant updates the rank per finished rollout [22].	Empirical only — the rank is selected on the same scores it thresholds. Reported alongside Bonferroni, not as the guaranteed claim.
censoring-aware upper bounds	One-sided bounds for doomed runs, where only $y >$ remaining budget is known [23–25].	One-sided upper coverage on the censored subset. $+\infty$ always valid but vacuous once the censored fraction exceeds α; the others empirical or assumption-dependent.
feasibility-gated intervals	A classifier on the same prefix features issues intervals only on predicted-feasible prefixes.	Coverage $\geq 1-\alpha$ conditional on passing the gate: selective, not marginal. Marginal coverage reported empirically.
stopping rule (τ)	Abort at the first turn whose feasibility score drops below τ, a conformal-risk-control gate [27] scored by Cost-of-Pass [26].	False-abort probability $\leq \delta$ per run, from rollout-level calibration on feasible runs.

C.2 Rollout-safe online evaluation

Online methods consume the test rows in their natural trajectory order, rollout by rollout and turn by turn. A prefix's label is the resource the run still needs to finish, which cannot be observed until the rollout ends, so a rollout's labels enter the update state only at its boundary and the state is frozen within a rollout (`conformal.aci_cqr`, `windowed_cqr`, `dtaci`, each taking a `groups` argument). The code rejects test streams whose rollout ids are not contiguous blocks: an id that reappears after a boundary would silently absorb a still-running rollout's labels and inflate coverage with no visible symptom. Revealing labels per prefix instead would use information a deployed guard does not have, so we report only the rollout-safe protocol.

C.3 Base predictor and features

The point and quantile predictors are LightGBM models [18] with `n_estimators` $= 200$, `learning_rate` $= 0.05$, `num_leaves` $= 15$ and `min_child_samples` $= 10$. CQR fits two quantile regressors, at $\alpha/2$ and $1-\alpha/2$; the absolute-residual baseline and the interval-driven stopping variant use a squared-error regressor; the feasibility gate is an LGBM classifier with the same settings. No hyperparameter search was run: these are the defaults fixed at the start of the project and used unchanged for every experiment, baseline included.

The feature matrix (`features.build_xy`) is: the turn index k; cumulative spend; burn rate (spend$/k$); the last, mean, standard deviation, minimum and maximum per-turn spend; the mean of the last three turns; and the difference between the last turn and the mean. When the agent emits its own interval, four columns are added — the two endpoints, the midpoint and the width — and a missing estimate is left as `NaN`, which LightGBM handles natively. Quantities known only at rollout end (final totals, success, feasibility) are listed in a leak set and asserted absent from the matrix at build time, so the exclusion is enforced rather than intended.

Table A2: **Full within-run results: every conformal method lands within 0.05 of the 0.90 target except the feature-free constant-quantile baseline on Search-R1, where the agent's own intervals miss by 0.39–0.67; windowed-CQR gives the narrowest bounded intervals on Sokoban and the second-narrowest on SWE-bench, while constant-quantile is sharpest on Search-R1.** Claude Opus 4.7, nominal 0.90, mean ± standard deviation over five rollout-level split seeds. Normalised width is interval width divided by the central 80% range of the labels (lower is sharper). "Unb." is the percentage of predictions with an infinite upper endpoint; width is shown as "–" where at least one seed produced a majority of unbounded intervals, making the across-seed summary infinite. n_{test} is test prefixes per seed. Bold marks, among bounded methods, the coverage closest to nominal and the narrowest width.

Environment	Method	Coverage	Norm. width (↓)	Unb. (%)	n_{test}
Sokoban	agent self-report	0.51	–	–	–
	constant-quantile	0.88 ± 0.05	1.10	0	87
	split-absres	**0.89 ± 0.03**	1.12	0	87
	split-CQR	0.92 ± 0.03	0.88	0	87
	conformalized-agent	0.93 ± 0.05	1.44	0	87
	guard+self-est	0.91 ± 0.04	0.91	0	87
	ACI-CQR	0.92 ± 0.02	0.88	2	87
	windowed-CQR	0.92 ± 0.02	**0.86**	0	87
	DtACI	0.92 ± 0.02	0.88	0	87
Search-R1	agent self-report	0.23	–	–	–
	constant-quantile	0.77 ± 0.17	**0.90**	0	24
	split-absres	0.94 ± 0.06	1.60	0	24
	split-CQR	0.90 ± 0.15	1.14	0	24
	conformalized-agent	**0.90 ± 0.10**	1.96	0	24
	guard+self-est	0.85 ± 0.17	1.07	0	24
	ACI-CQR	0.90 ± 0.15	1.14	0	24
	windowed-CQR	0.92 ± 0.10	1.10	0	24
	DtACI	0.94 ± 0.06	–	16	24
SWE-bench	agent self-report	0.35	–	–	–
	constant-quantile	**0.92 ± 0.03**	2.87	0	161
	split-absres	0.95 ± 0.09	2.56	0	161
	split-CQR	0.96 ± 0.10	2.11	0	161
	conformalized-agent	0.97 ± 0.07	2.23	0	161
	guard+self-est	0.96 ± 0.09	1.97	0	161
	ACI-CQR	0.94 ± 0.09	–	14	161
	windowed-CQR	0.87 ± 0.08	1.29	0	161
	DtACI	0.87 ± 0.07	**1.22**	0	161

C.4 Conformal methods and their settings

Every method uses the finite-sample corrected quantile: with n calibration scores the level is $\lceil (n+1)(1-\alpha) \rceil / n$, read with higher interpolation, returning $+\infty$ when that level exceeds 1. All intervals are clipped below at zero, since a remaining budget cannot be negative. $\alpha = 0.10$ throughout.

split-CQR uses the score $\max(\hat{f}_{\text{lo}} - y,\ y - \hat{f}_{\text{hi}})$ and one global $\hat{q}$; it is the offline default. **Mondrian-CQR** applies the same score with one $\hat{q}$ per turn-depth bucket, split at $k = 1$, $k = 2$ and $k \geq 3$; a bucket holding fewer than 10 calibration scores falls back to the global $\hat{q}$.

ACI updates the working level by $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$. γ is selected on calibration data only: the calibration rollouts are split in half, ACI is streamed over the second half under the same rollout-safe revelation used at test time, and the $\gamma \in \{0.005, 0.01, 0.02, 0.05\}$ whose coverage lands closest to $1 - \alpha$ is kept. One deliberate deviation from canonical ACI: where the canonical algorithm emits the empty set once $\alpha_t \geq 1$, we clamp the quantile lookup at 0.999 and emit a finite interval. The deviation is conservative in both coverage and the error feedback, and we state it rather than diverge silently.

windowed-CQR keeps a rolling window of the 100 most recent conformity scores, warm-started from calibration, and reads the interval half-width off that window's empirical quantile, so the score distribution itself tracks drift rather than reweighting a frozen calibration set. Earlier drafts of this

Table B1: **Breadth across all 19 token-budget runs: windowed-CQR clears 0.85 on 18 of them and 0.90 on 11, while the agents' own intervals clear neither anywhere.** Marginal coverage against nominal 0.90 on a single rollout-level split (seed 0), so the Claude Opus 4.7 rows differ slightly from Table A2, which averages five seeds. Four decimals are shown on the three guard columns so the two counts above are directly checkable off the table; at three decimals SWE-bench/Gemini DtACI rounds to 0.850 and appears to clear 0.85 when it does not. "–" marks a run whose records carry no agent interval. †Warehouse is listed for dataset composition only; its 0.424 is joint coverage across three resources (Table 2), not a token-budget marginal.

Environment	Model	Rollouts	Prefixes	Agent	coverage, closest to nominal 0.90 in bold		
					split-CQR	windowed-CQR	DtACI
Sokoban	Claude Opus 4.7	123	355	0.513	0.8667	**0.9000**	**0.9000**
	Claude Opus 4.6	116	410	0.446	0.8077	0.8750	**0.8846**
	Claude Sonnet 4.6	126	400	0.524	0.9623	**0.9151**	0.9528
	GPT-5.2-instant	128	661	0.481	0.8176	**0.8986**	0.8716
	GPT-5.2-Codex	128	592	–	0.9933	0.9600	**0.9467**
	Gemini 3.1 Pro	126	389	0.125	0.9425	**0.9310**	**0.9310**
	Qwen3-235B	126	1,038	0.090	0.9268	**0.9024**	0.8862
Search-R1	Claude Opus 4.7	66	87	0.233	**1.0000**	**1.0000**	**1.0000**
	Claude Opus 4.6	52	90	0.478	**0.9130**	**0.9130**	**0.9130**
	Claude Sonnet 4.6	64	102	0.386	1.0000	**0.9200**	1.0000
	GPT-5.2-instant	125	324	0.198	0.8256	**0.8605**	0.8488
	GPT-5.2-Codex	126	262	0.202	0.9649	0.9298	**0.8772**
	Gemini 3.1 Pro	83	85	0.202	0.8182	**0.8636**	0.8182
	Qwen3-235B	124	1,079	0.025	0.9259	0.9259	**0.8765**
SWE-bench	Claude Opus 4.7	64	712	0.354	0.9926	0.9185	**0.9037**
	Claude Sonnet 4.6	64	1,258	0.251	0.9101	0.8705	**0.8957**
	GPT-5.2-instant	64	1,313	0.385	**0.8840**	0.8589	0.8370
	Gemini 3.1 Pro	64	2,327	0.254	0.8497	**0.8899**	0.8497
	Qwen3-235B	60	3,715	0.034	1.0000	0.7727	**0.8271**
Warehouse	Opus 4.7 + 4 swap targets	64 ea.	1,280 ea.	0.424†	–	–	–

Table B2: **Marginal coverage hides a first-turn deficit on Sokoban, and Mondrian calibration is what removes it.** Coverage by turn-depth bucket, nominal 0.90, mean over five seeds. Search-R1 rollouts are short enough that no $k \geq 3$ bucket exists.

Environment	Method	Marginal	$k = 1$	$k = 2$	$k \geq 3$
Coverage against nominal 0.90; the value closer to nominal in each column is bold.					
Sokoban	split-CQR	**0.92**	0.86	**0.93**	0.96
	Mondrian-CQR	0.94	**0.93**	0.96	**0.94**
Search-R1	split-CQR	**0.90**	**0.91**	**0.89**	–
	Mondrian-CQR	0.89	**0.89**	**0.89**	–
SWE-bench	split-CQR	**0.96**	**0.99**	**0.99**	**0.95**
	Mondrian-CQR	0.97	**0.99**	**0.99**	0.96

work called the method SPCI; that overclaims, because SPCI fits a quantile regression on the residual series. SPCI [19] is the inspiration, not the implemented method.

DtACI aggregates ACI experts at $\gamma \in \{0.005, 0.01, 0.02, 0.05, 0.1, 0.2\}$ with learning rate $\eta = 2.0$ and uniform mixing $\sigma = 0.01$. After each revealed label, experts are scored by pinball loss against β_t, the largest miscoverage level whose quantile would still have covered the observed score, and the working level is the weight-averaged α_t.

C.5 Baselines

Seven comparators run on the same splits and seeds as the guard.

Table B3: **The feasibility gate trades retained volume for coverage on what it keeps: abstaining on predicted-doomed prefixes holds 0.92–0.95 coverage on capacity, where marginal calibration reaches only 0.89.** Warehouse, mean over five seeds. "Kept" is the fraction of prefixes the gate admits, "caught" the fraction of doomed prefixes it correctly abstains on, "false abort" the fraction of feasible prefixes it wrongly rejects, and "coverage" the interval coverage among admitted feasible prefixes. Time binds in only 30% of censored runs and the gate catches almost all of them, so its row is flat. δ here is the gate's prefix-level tolerance, not the rollout-level false-abort tolerance of §5.5.

Resource	δ	Kept (↑)	Caught (↑)	False abort (↓)	Coverage (↑)
time (weeks)	0.02	0.88	0.93	0.00	1.00
	0.05	0.88	0.93	0.00	1.00
	0.10	0.88	0.93	0.00	1.00
	0.20	0.88	0.93	0.00	1.00
	0.30	0.88	0.93	0.00	1.00
capacity (item-weeks)	0.02	0.89	0.31	0.04	0.95
	0.05	0.78	0.64	0.05	0.95
	0.10	0.70	0.74	0.07	0.95
	0.20	0.57	0.82	0.19	0.95
	0.30	0.51	0.85	0.23	0.94
cost (USD)	0.02	0.86	0.60	0.05	0.92
	0.05	0.81	0.69	0.10	0.92
	0.10	0.80	0.72	0.10	0.92
	0.20	0.74	0.75	0.21	0.93
	0.30	0.74	0.76	0.21	0.93

1. **Agent self-report.** BAGEN's own per-sample indicators of whether the agent's interval contained the truth, read from the official estimation records. No calibration is applied. This is the uncalibrated baseline the paper is written against.
2. **Constant-quantile.** Feature-free: the empirical $[\alpha/2,\ 1-\alpha/2]$ interval of the calibration labels, identical at every test prefix. If the guard barely beats this, the features are carrying nothing.
3. **Split-absres.** A LightGBM point prediction plus or minus the calibrated absolute-residual quantile.
4. **Conformalized-agent.** The agent's own interval midpoint used as the point predictor and conformalized on $|y - \mathrm{mid}|$, which requires no training at all.
5. **Guard+self-est.** split-CQR on the guard features augmented with the agent's estimate columns; an ablation on whether the self-report adds anything the usage stream does not already carry. Its answer is reported in §5.1: it does not.
6. **Frozen split-CQR.** split-CQR calibrated on the source environment or model and applied unchanged to the target stream — the comparison the online methods are measured against in §5.2.
7. **Hard caps.** Abort once spend so far exceeds a fixed cap. Dollar caps are swept at the 20th, 35th, 50th, 65th and 80th percentiles of realised run totals, and week caps at $0.5\times$ the 25th, $0.75\times$ the 50th and $1.0\times$ the 75th percentile of the task's time budget — three caps, not nine.

Every baseline sees the same training split, the same calibration split, the same test stream and the same seeds as the guard, and the two that fit a model use identical LightGBM settings; none is handicapped relative to the guard. BAGEN's SFT+RL estimator is not included as a baseline: it is a fine-tuned agent rather than an external wrapper, its reported numbers come from a different evaluation protocol, and re-running it would require training the underlying LLM, which is exactly what this work avoids. We quote it in §2 as context and do not place it in any results table.

C.6 Metrics

Coverage is the fraction of test prefixes whose interval contains the true remaining requirement. Normalised width is the interval width divided by the central 80% range of the labels, $q_{0.9} - q_{0.1}$,

Table B4: **Dropping censored runs under-covers exactly the expensive failures, and every informative repair trades validity for sharpness.** One-sided upper-bound coverage on Warehouse at nominal 0.90, split by whether the resource was the binding constraint. $+\infty$ is always valid and always vacuous; "drop" is the naive comparison. Kaplan–Meier is a diagnostic, not a calibration option. Bold marks the highest censored-binding coverage among the four treatments that compete on it, excluding the vacuous $+\infty$ and the diagnostic. Coverage for capacity and cost is measured against an interpolated label (12–27% median relative error), not ground truth.

Resource	Treatment	Feasible	Censored		Binding
		(↑0.90)	binding	non-binding	fraction
time (weeks)	drop (naive)	1.00	**1.00**	1.00	0.30
	$+\infty$	1.00	1.00	1.00	0.30
	substituted bound	1.00	**1.00**	1.00	0.30
	IPCW	0.96	0.96	0.96	0.30
	per-resource bound	0.90	0.90	0.90	0.30
	Kaplan–Meier (diagnostic)	1.00	1.00	1.00	0.30
capacity (item-weeks)	drop (naive)	0.94	0.77	0.98	0.34
	$+\infty$	1.00	1.00	1.00	0.34
	substituted bound	0.99	**0.98**	1.00	0.34
	IPCW	0.93	0.75	0.98	0.34
	per-resource bound	0.91	0.74	0.98	0.34
	Kaplan–Meier (diagnostic)	0.94	0.78	0.99	0.34
cost (USD)	drop (naive)	0.88	0.50	0.97	0.42
	$+\infty$	1.00	1.00	1.00	0.42
	substituted bound	1.00	**0.99**	1.00	0.42
	IPCW	0.88	0.50	0.96	0.42
	per-resource bound	0.86	0.46	0.94	0.42
	Kaplan–Meier (diagnostic)	0.92	0.58	0.98	0.42

which makes environments whose token scales differ by orders of magnitude comparable; the median width is used wherever a method can emit unbounded intervals, and the across-seed summary is reported as "–" where at least one seed's median is itself infinite. Unbounded fraction is the share of predictions with an infinite upper endpoint, and it appears beside every coverage number, because an unbounded interval covers everything and informs nothing.

Confidence intervals come from a cluster bootstrap that resamples whole rollouts rather than prefixes, with $B = 500$ resamples; per-sample binomial intervals are too tight here, because the prefixes of one rollout are correlated. Across seeds, the per-seed bootstrap draws are pooled and the 5th and 95th percentiles of the pooled distribution are reported as a 90% interval; these are recorded in the released result tables rather than plotted. Averaging per-seed interval endpoints, which an earlier revision of this work did, is not a confidence interval at any level: a seed whose coverage is 1.00 contributes a degenerate $[1, 1]$ whose mean swallows the seeds that landed near 0.70. The $\pm$ shown in Table A2 and the error bars in Figure 2 are both the standard deviation across the five seeds. The pooled bootstrap intervals are not plotted; they are carried in the released result tables.

For the stopping rule, the false-abort rate is the fraction of *feasible* test rollouts — those that would have finished within budget — that were stopped, and the doomed-savings fraction is the spend avoided on rollouts that could not finish divided by the spend those rollouts would otherwise have incurred. Cost-of-Pass is total spend under a policy divided by the number of runs that still pass [26]. All economic results in this paper are Warehouse, which is priced natively in simulated dollars; no token-environment economics are reported.

C.7 Multi-resource and censored calibration

Warehouse binds time in weeks, capacity in item-weeks and cost in simulated dollars at once, and per-prefix ground truth for all three comes from the official estimation records rather than from dialogue extraction. Independent calibration sets each margin at α; Bonferroni sets each at $\alpha/3$, which restores a joint guarantee of $1 - \alpha$ by the union bound and is the headline multi-resource method. The empirical-copula variant shares one rank across resources and is tighter, but that rank is selected on the same calibration scores it then thresholds, so it carries no guarantee and is reported as

an observation only. Under a model swap, an adaptive variant recomputes both the margins and the shared rank from a rolling window of recent three-resource error vectors, updated only once a rollout has finished; as Table 2 shows, it is the only scheme that stays near nominal under that shift, and it is precisely the one with no guarantee.

For censored runs the one-sided upper bound is calibrated five ways (`censored.upper_qhat`): *drop*, which uses uncensored scores only and is the naive comparison; *infinity*, which treats censored scores as $+\infty$ and is always valid but vacuous once the censored fraction exceeds α; *bound*, which substitutes each censored score's known lower bound; *ipcw*, an inverse-probability-of-censoring-weighted quantile using the feasibility classifier's probability; and *resource_bound*, which charges a censored run only against the resource that actually stopped it, since a run censored by its time budget says nothing about its remaining cost. A Kaplan–Meier estimator [25] is a diagnostic rather than a sixth option: it returns $+\infty$ exactly when the upper tail is entirely censored, which is the rigorous statement that no finite valid bound exists. Table B4 reports the five calibrations together with the Kaplan–Meier diagnostic.

One caveat governs how the censored numbers should be read. Warehouse's censored rows retain their realised totals, so a label can be masked, calibrated as if censored, and then measured against the recovered value. That recovery is exact only for time in weeks; for cost and capacity it inherits a week-proportional interpolation whose relative error runs at a median of 12–27% on comparable uncensored rows. Censored-stratum coverage for those two resources is therefore coverage against an approximate label, not against ground truth.

The feasibility gate is an LGBM classifier on the same prefix features. It issues intervals only on predicted-feasible prefixes, so the resulting guarantee is selective — coverage conditional on passing the gate — and marginal coverage over all prefixes is reported as an empirical observation. The gate is swept at its own prefix-level tolerance $\delta_{\text{gate}} \in \{0.02, 0.05, 0.10, 0.20, 0.30\}$, which is a different quantity from the rollout-level false-abort tolerance δ of Appendix C.8; Table B3 reports the full sweep.

C.8 Stopping rule and economics

The reported rule aborts at the first turn whose feasibility score falls below τ. Because the rule aborts a rollout exactly when that rollout's minimum prefix score crosses τ, τ is calibrated at the rollout level: it is set to the score at rank $\lfloor (n+1)\delta \rfloor$ among the sorted minimum scores of the n feasible calibration rollouts, which bounds by δ the probability of aborting a run that would have finished. This is a per-run guarantee matched to the sequential decision, not a per-prefix statement, and a δ small enough to give rank 0 aborts nothing at all — which is why the $\delta = 0.02$ and $\delta = 0.05$ points in Figure 4 save nothing. δ is swept over $\{0.02, 0.05, 0.10, 0.15, 0.20, 0.30, 0.40\}$ for the frontier; $\delta = 0.10$ is the operating point quoted in §5.5 and the abstract.

An interval-driven variant, which aborts when spend so far plus the calibrated lower bound exceeds the budget, is implemented in the released code and calibrated the same way at rollout level. It is not the rule behind the reported economics, and we do not claim the two are equivalent.

The economic numbers are counterfactual replays, not live measurements: each completed run is truncated where the guard would have aborted it and the avoided spend is summed. This is fair in that every run's real outcome and cost are known, but it assumes that adding the guard would not change how the agent behaves and that the logged sample is representative. §6 states the limitation; it is repeated here so the numbers are not read as a deployment result.

C.9 Data, environment, and reproduction

All experiments use logged rollouts from BAGEN [13], fetched from its public Hugging Face release at the pinned revision `58189576e54b675fdd0e1d6c1c9f189c2992732f`. The dataset card states no licence, so we use the data as publicly released for research and credit the originators rather than asserting terms the release does not state. Pinning matters because the upstream repository is live: without it, a re-run silently evaluates different data. The extractor reproduces BAGEN's turn-excluding-history token accounting exactly, with 0 of 355 mismatches against the official per-sample ground truth on the reference run. We release no derived copy of the data; the fetch script downloads it from the upstream repository. That script and the rest of the pipeline are not attached to

this submission: the implementation will be released publicly under an MIT licence on acceptance. This appendix is written to stand without it.

The whole pipeline runs inside a container whose base image is pinned by digest (`python:3.12-slim-bookworm`), with numpy 2.5.1, pandas 3.0.3, LightGBM 4.6.0, scikit-learn 1.9.0, scipy 1.18.0 and matplotlib 3.11.0 resolved from a hash-locked requirements file. LightGBM is MIT-licensed; numpy, pandas, scipy and scikit-learn are BSD-3-Clause; matplotlib uses a PSF-style licence. Our own code is released under MIT. The guard is CPU-only: no GPU, no model weights, and no access to logits or environment internals, since every feature comes from the usage accounting that provider APIs already report. No LLM is trained, fine-tuned or queried by us at any stage — all agent rollouts are pre-existing logged data — so the compute cost of this work is the cost of fitting gradient-boosted trees on tables of at most a few thousand rows. Reported numbers reproduce bit-identically across host architectures apart from interval widths, which can differ by one unit in the last place (2.2×10^{-16}).

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Justification: Appendix C specifies everything needed to rebuild the pipeline: the by-rollout 50/25/25 split with seeds 0–4 and the exact RNG (C.1), the rollout-safe label-revelation protocol, the LightGBM settings and the complete feature list with its enforced leak set, the finite-sample corrected quantile and the settings of every conformal variant including window length, ACI step-size selection and the DtACI expert grid, all seven baselines (C.5), the definition of every reported metric and the cluster bootstrap (C.6), the joint and censored calibration variants (C.7), and the stopping-rule calibration and δ sweep (C.8). Appendix C.9 pins the input data to a single dataset revision and the software environment to a digest-pinned container with a hash-locked requirements file.

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Justification: The data is fully open: every input is a public BAGEN logged rollout, fetched at the pinned revision recorded in Appendix C.9 by a download script, so anyone can obtain exactly the inputs we used. The code is not attached to this submission. We will release the full pipeline publicly under an MIT licence on acceptance, together with the digest-pinned Dockerfile, the hash-locked requirements file, and the single entry point that regenerates every figure and table in the paper and machine-verifies 11 headline claims and all 35 quantitative claims against what it just produced, exiting non-zero on any mismatch. In the meantime Appendix C specifies the split procedure, every hyperparameter, every baseline, every metric formula and the exact software versions, which is what an independent reimplementation needs.

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Justification: Section 4 gives the environments and their budgets (Table 1), the 50/25/25 by-rollout split, the five seeds, the three shift scenarios, the seven agents and the resulting 19 model×environment runs, and the comparator list. Appendix C gives the rest: the split procedure, the LightGBM hyperparameters shared by the quantile regressors, the absolute-residual baseline and the feasibility classifier, the full feature matrix, $\alpha = 0.10$ throughout, and the exact sweep grids for the hard-cap baselines, the feasibility gate and δ. No hyperparameter search was run; the settings are project defaults used unchanged for the guard and for every baseline.

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    Justification: No human subjects were involved at any stage, so no IRB review or equivalent approval was required. The study consists entirely of statistical calibration applied to a previously released public dataset of machine-generated agent trajectories.

    Guidelines:

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    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

    Answer: [N/A]

    Justification: No LLM is a component of the core method. Budget Guard requires no LLM fine-tuning and is model-agnostic: its predictors are gradient-boosted trees over usage-accounting features, and it reads no weights, logits or model outputs beyond the agent's

optional self-reported interval (Section 3, Appendix C). We neither train, fine-tune nor query any LLM in this work; the seven agents named in Section 4 are the sources of the pre-existing logged rollouts we analyse, not components of the method.

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